



Pharm*Achieve*

WARFARIN DOSING

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

BID Twice Daily

CYP Cytochrome P450

DVT Deep Vein Thrombosis

HS Hora Somni (At Bedtime)

INR International Normalized Ratio

LMWH Low Molecular Weight Heparin

NSAID Non-Steroidal Anti-Inflammatory Drug

PE Pulmonary Embolism

PO Per Os (Oral)

POD Post-Op Day

POD-1 Post-Op Day -1 (day before surgery)

PPI Proton Pump Inhibitor

TIA Transient Ischemic Attack

VAS Visual Analog Scale

VKOR Vitamin K Epoxide Reductase

VTE Venous Thromboembolism

INTRODUCTION TO WARFARIN

Indirect anticoagulant

- Decreases ability of liver to produce coagulation factors II,VII,IX,X
- Decreases ability of liver to produce endogenous anticoagulants, such as protein C and S

Common indications

- Stroke prevention in atrial fibrillation
- Treatment of VTE
- Long-term secondary prevention of VTE
- VTE prevention in patients with mechanical heart valves

WARFARIN DOSING

Maintenance dose varies

- Anywhere from <1 mg to >20 mg daily



Reasonable starting dose is 5 mg PO daily

- If frail, or underweight, or of Asian descent: 1-2 mg PO daily
- Initiation with LMWH x 5 days and until INR is 2 x 2 days



Factors that affect the maintenance dose

- Age, weight, race, diet, genetic variation in VKOR (enzyme implicated in warfarin action), genetic variation in CYP2C9 (enzyme that metabolizes warfarin), drug interactions, alcohol, other disease states, level of daily activity

MONITORING

Routine lab monitoring required for all warfarin patients

Target INR generally between 2.0-3.0

- Exception: mechanical mitral valve (2.5-3.5)

Timeframe for INR change normally takes 3-7 days depending on individual patient factors

- Monitoring should be done no earlier than 2-3 days after initial dose or dose adjustments
- Once stable, frequency can be reduced to once monthly (every 12 weeks in very stable patients)

INTERACTIONS

Antiplatelet Agents

Risk of bleeding is almost doubled when used concomitantly

NSAIDs

May use celecoxib or traditional NSAID+PPI to mitigate gastrointestinal bleeding risk

Antibiotic

Risk of increased INR

Acute Infection

More frequent monitoring may be necessary

Alcohol

May alter INR (both increased bleed risk or decreased anticoagulant effect possible)

Supplements

- St. John's Wort – decreased anticoagulant effects
- Gingko Biloba – increased bleeding risk

WARFARIN REVERSAL

Vitamin K (PO)

- 2.5-10 mg – measure INR after 12-48 hours and administer subsequent dose as needed
- Following PO administration, reversal expected in 6-10 hours, peak effect within 24-48 hours

Emergent reversal: IV vitamin K *and* Octaplex® or Beriplex® (four-factor prothrombin complex)

- Weight-based dosing (units/kg) based on INR value or fixed – measure INR at baseline and 30 minutes post-dose
- Significant INR decline within 30 minutes, maintained for up to ≥ 24 hours

APPROACH TO OUT-OF-RANGE INR

Questions to consider

- **Missed any doses in the past week?**
Use strategies to improve compliance (i.e. blister packs, phone reminders, dosing calendar)
- **Started or stopped any medications/supplements recently?**
Antibiotics, azole antifungals, levothyroxine dose changes can increase INR
- **Changes in diet recently?**
Vitamin K rich foods (green leafy vegetables, soy, avocado)
- **Changes in alcohol intake recently?**
More than 2 drinks will transiently increase INR
- **Current fever, diarrhea, flu, recent cold?**
This can cause higher than normal INR value

Determine whether a single dose change or maintenance dose change is required:

Single dose change if slightly out-of-range INR (0.5 above/below target) and transient cause identified

- Make a one-time dose change (increase or hold by $\frac{1}{2}$ -1 dose) and resume current dose OR continue current dose
- Repeat INR in 1-2 weeks

Change in maintenance dose considered if 2 or more out-of-range INR trending in the same direction



See next slide for management approach

WARFARIN DOSE ADJUSTMENTS



Target INR 2.0-3.0

INR	INR <2.0	INR 3.1-3.5	INR 3.6-4.0	INR 4.1-8.9**	INR >9.0**
Dose	Increase by 10-15%*	Decrease by 0-10%	Hold 0-1 dose, decrease by 10-15%	Hold 0-2 doses, decrease by 10-15%	Hold 2 doses, decrease by 15-20%
Repeat	Repeat INR within 1 week	Repeat INR within 2 weeks	Repeat INR within 1 week	Repeat INR in 2 days	Repeat INR next day



* Consider 15% increase if INR ≤ 1.5 without explanation

** If INR > 4.5 and < 10, in the absence of clinically significant bleeding: temporary holding of warfarin doses is sufficient; no need to add vitamin K

If INR > 10, vitamin K can be considered even in the absence of bleeding, based on individual risk factors (e.g. bleeding risk, thrombosis risk, ease of repeat INR testing)

PERIOPERATIVE WARFARIN MANAGEMENT

IS BRIDGING REQUIRED?

If high thromboembolic risk, bridging is recommended:

Mechanical prosthetic mitral valve, older generation mechanical aortic valve, CHADS₂ = 5 or 6, recent (within last 3 months) stroke/TIA/DVT/PE, prior VTE during interruption of warfarin therapy

If intermediate thromboembolic risk, bridging is optional and is assessed on a case-by-case basis: CHADS₂ = 3 or 4, newer generation mechanical aortic valve, VTE within last 3-12 months

If bridging is required:



Stop warfarin 5 days pre-op and obtain INR value on POD-1, use therapeutic LMWH for approximately 3 days pre-op and administer in morning (day prior to surgery, half-dose LMWH)

If no bridging required:



Stop warfarin 5 days pre-op and obtain INR value on POD-1

CASE 1

Mrs. TB is an 82-year-old female who has been on warfarin for 20 years. She has been suffering from increasing joint pain recently and has been scheduled for a total knee replacement surgery next month. TB has a past medical history significant for ischemic stroke, osteoarthritis, type 2 diabetes mellitus, hypertension, and atrial fibrillation. She takes acetaminophen 1000 mg PO BID, metformin 1000 mg PO BID, warfarin 4 mg PO daily, and amlodipine 5 mg PO HS. She has no known drug allergies. Her blood pressure is recorded today as 125/80 mmHg, her CHADS₂ score is 5, VAS = 7/10 and INR is 2.6.

1. Which of the following would be an appropriate recommendation for Mrs. TB pre-op?
 - a) Stop warfarin 2 days pre-op and bridge with therapeutic enoxaparin until 24 hours pre-op
 - b) Stop warfarin 5 days pre-op and bridge with therapeutic enoxaparin until the day of surgery
 - c) Stop warfarin 5 days pre-op and bridge with therapeutic enoxaparin, with half the dose administered on the day prior to surgery
 - d) Stop warfarin 5 days pre-op with no bridging



CASE 1

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2. It has been two months since Mrs. TB's surgery. She is very pleased and seems to be in much less pain than prior to her operation. She visits her family physician for a routine follow-up and her INR comes back as 1.7. Which of the following could be a reason for this?
- a) Recently treated with a course of nitrofurantoin for a suspected UTI
 - b) Recently started St. John's Wort to help with her low mood
 - c) Recently celebrated her 83rd birthday and had 3-4 drinks over the weekend
 - d) Stopped eating foods rich in vitamin K after reading an article her sister sent to her



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3. Mrs. TB has had two out-of-range INRs over the last few weeks; one came back as 1.7 and the second as 1.6. Which of the following is appropriate in this case?
- a) Increase total weekly dose by 10% and repeat INR in 3 weeks
 - b) Increase total daily dose by 10% and repeat INR in 1 week
 - c) Increase total weekly dose by 15% and repeat INR in 1 week
 - d) Increase total daily dose by 5% and repeat INR in 2 weeks



REFERENCES

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CHANGE LOG

October 2024

- Slide 5: In the maintenance dosing section, added 'greater than' in front of 20 mg as per Thrombosis Canada guideline
- Slide 7: Updated drug interaction information provided for alcohol. Changed from 'may decrease serum warfarin levels' to altered INR levels as both increases/decreases possible
- Slide 8: Added Beriplex® as another example of a four-factor prothrombin complex
- Slide 12-15: Changed Tylenol to acetaminophen, and added that patient has had a prior stroke to correlate with CHADS2 score of 5

July 2025

- Formatting, grammar changes made throughout; legend and references updated
- Slide 9: Updated dosing and pharmacokinetic information, removed SL
- Slide 12: Updated bridging therapy requirements

September 2025

- Slide 3: Added POD-1 to legend
- Slide 13: Modified option C

January 2026

- Slide 2: Updated NAPRA competencies
- Slide 8: Capitalized AND in emergent reversal to emphasize that IV vitamin K is administered *along with* a four-factor prothrombin complex (Octaplex® or Beriplex®)
- Slide 10: Added information on when to consider vitamin K based on INR ranges and bleeding risk
- Slide 15: Updated references